BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022 • Paper Code: 9221

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** 9221

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. The least count of vernier calliper is ..... cm.

(A) 0.0001 (B) 0.001

(C) 0.01 (D) 0.1

Q2. The slope of distance time graph represents .....

(A) displacement (B) speed

(C) acceleration (D) distance

Q3. The torque in uniformly rotating fan having blade of length of 0.5 m is .....

(A) 0.5 Nm (B) −0.5 Nm

(C) 0 Nm (D) 0.1 Nm

Q4. The first condition of equilibrium is .....

(A) $\Sigma F=0$ (B) $\Sigma \tau=0$

(C) $\Sigma P=0$ (D) $\Sigma W=0$

Q5. An object at earth and taken to moon should have .....

(A) same mass/less weight (B) same mass/more weight

(C) less mass/less weight (D) less mass/same weight

Q6. The angle at which horizontal and vertical components of force are equal is .....

(A) 30° (B) 45°

(C) 60° (D) 90°

Q7. An object of mass 10 kg is lifted vertically through a height of 5 m. The gravitational potential energy gained by the object is .....

(A) 0.5 J
(B) 2 J
(C) 50 J
(D) 490 J

Q8. The unit used for pressure in weather map is .....

(A) Pa
(B) bar
(C) Atm
(D) Nm^{-2}

Q9. The specific heat capacity of water is ..... J $kg^{-1}K^{-1}$.

(A) 5190
(B) 4360
(C) 0
(D) 4190

Q10. The expression used to measure Young's Modulus is .....

(A) F/Ax
(B) Fl/A
(C) Fl/Ax
(D) Ax/Fl

Q11. The transfer of heat from the sun on earth is due to .....

(A) radiation
(B) convection
(C) conduction
(D) absorption

Q12. The fahrenheit and celsius scale reading of temperature coincide at .....

(A) $-40°$
(B) $-460°$
(C) $-273°$
(D) $+273°$

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 32 | **Paper Code:** 9221

Q1. Attempt any eight of the following short questions. Each question carries 4 marks

(i) Differentiate between base and derived physical quantities.

(ii) Describe a situation in which the speed of an object is constant while the velocity is not.

(iii) Which law of Newton is called law of inertia and why?

(iv) Why do tightrope walkers carry a long narrow rod?

(v) What would happen to your weight on earth if the mass of the earth doubled but its radius stayed the same?

(vi) Moon is attracted by earth, why it does not fall on earth.

(vii) Temperature of an object is 250 K. Find this temperature in centigrade and fahrenheit scale.

(viii) Why is the freezer compartment kept at the top of a refrigerator? Discuss.

(ix) Why are white clothes preferred in summer and black in winter?

(x) What is elasticity? Why is the steel more elastic than rubber?

(xi) Why are the small gaps left behind the girders mounted in wall.

SECTION-C

Maximum Marks: 21

> Note: Attempt any three of the following questions. Each question carries 7 marks

Q2.

(i) Graphically prove the first and second equations of motion.

(ii) With what speed must a ball be thrown vertically from ground level to rise to a maximum height of 50 m.

Q3.

(i) What is resolution of a force? How can a force be resolved into its perpendicular components?

(ii) At which altitude above earth's surface would the gravitational acceleration be 4.9 ms^{-2}?

Q4.

(i) Why does liquid exert pressure? Derive an expression for the pressure exerted by the liquid.

(ii) A submarine with a pressure 1.6×10^{7} Pa and density of sea water is 1025 kgm^{-3}. How deep may this submarine descend in the ocean?

Q5.

(i) What is kinetic energy? Prove that kinetic energy of a body is K.E.=$\frac{1}{2}mv^2$.

(ii) A bullet of mass 50 g fired from a gun has kinetic energy 3125 J. Find its velocity.